

CJ Panganiban Conversational AI Pipeline Review

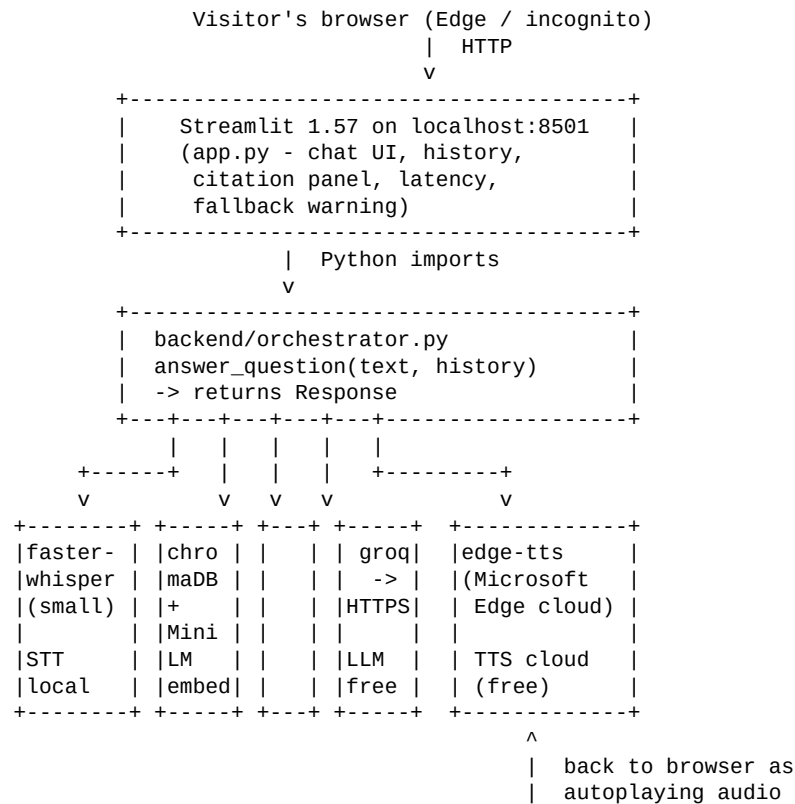
As of 2026-05-07

Supervaise × Foundation for Liberty and Prosperity

Phase A · all-free dev stack on the laptop · pre-OpenAI cutover

1. High-level architecture (Phase A — May 30 demo)

Everything runs on the dev laptop except the Groq LLM call. Streamlit imports the orchestrator directly (no HTTP layer).



2. Frontend layer

Item	Value
Framework	Streamlit 1.57.0
Entry point	app.py
URL	http://localhost:8501
Chat primitives	st.chat_input, st.chat_message, st.audio_input, st.audio
Conversation history	st.session_state.history (list of {role, content, meta})
Citation rendering	_render_meta() — bucket, title, date, distance, URL, unsafe-tag
File watcher	DISABLED via .streamlit/config.toml (Windows watchdog crash fix)
Telemetry	Disabled (gatherUsageStats = false)

Visitor flow: record (or type) → see “Heard:” caption → see “Thinking...” spinner → see assistant message + citations + latency + (optional) fallback warning → hear voice playback.

3. STT (Speech-to-Text)

Item	Value
Library	faster-whisper 1.2.1
Wrapper	backend/stt.py
Model	small (just bumped from base)
Disk size	~470 MB
Quantization	int8 (CPU-friendly, smaller memory footprint)
Device	cpu
Decode params	beam_size=1 (fastest greedy decoding)
Cache location	./models/faster-whisper-small/ (gitignored)
Download mechanism	huggingface_hub.snapshot_download with local_dir
Latency	~3-5 sec for a 5-15 sec audio clip on a typical laptop CPU
Languages	Auto-detect; multilingual; output is English (Taglish input fine)
Configurable via	WHISPER_MODEL env var (could go up to 'medium' if needed)

4. LLM (chat completion)

Item	Value
Provider	Groq (free tier)
Library	groq 1.2.0
Wrapper	backend/llm.py — single chat(messages) function
Model	llama-3.3-70b-versatile
Where it runs	Groq's cloud servers (only remote hop in the pipeline)
API key env var	GROQ_API_KEY
Model env var	GROQ_MODEL
Latency	~1-2 sec typical; can spike to 3-5 sec under load
Phase B target	OpenAI GPT-4o (single config flip)

5. TTS (Text-to-Speech)

Item	Value
Library	edge-tts 7.2.8 (Microsoft Edge's TTS endpoint)
Wrapper	backend/tts.py — synthesize(text, voice) returns mp3 bytes
Voice	en-US-AndrewNeural (US male, configurable via TTS_VOICE)
Other male options	en-US-GuyNeural, en-US-EricNeural, en-GB-RyanNeural, en-AU-WilliamNeural
Cost	Free, no API key required
Latency	~1-2 sec for a 100-150 word answer

Item	Value
Implementation note	Async-only API; wrapped with <code>asyncio.run()</code> per call
Caching	@st.cache_data on synthesize so same text doesn't re-synth

6. Persona & fallback (the LLM's instruction layer)

Item	Value
System prompt file	prompts/instructions.txt
Loader	backend/prompts.py — <code>system_prompt()</code> returns persona + substituted fallback
Persona rules (6)	(1) source-grounded only; (2) 100-150 word output cap; (3) English only (Taglish input OK); (4) measur
Fallback file	prompts/fallback.txt — 5 variants separated by '---'
Variant selection	Random per call (<code>random.choice</code>) — different phrasings per visitor
Owner of final wording	Jacob Barbosa (FLP) — current text is placeholder per OD-1, OD-3, OD-8

The persona prompt is injected as a *system* message prepended to every Groq call.

7. Embeddings layer

Item	Value
Library	sentence-transformers 5.4.1
Model	all-MiniLM-L6-v2 (changed from all-mpnet-base-v2 with v2 spec)
Dimensions	384 (was 768 before v2 handover)
Disk size	~90 MB
Where it runs	Local laptop CPU
Used in two places	(a) ingestion-time chunk embedding in <code>ingest_columns.py</code> ; (b) query-time question embedding in <code>backend/retrieval.py</code>
Cache	~/.cache/huggingface/hub/ (downloaded on first use; copy-mode on Windows)
Phase B target	OpenAI text-embedding-3-small (1536-dim) — REQUIRES re-embedding all chunks

8. Vector store

Item	Value
Engine	chromadb 1.5.8 (embedded, in-process, SQLite-backed)
Persist directory	./chroma_store/ (gitignored)
Collection name	cjp_columns_dev
Distance metric	cosine
Index	HNSW (Chroma default)
Total chunks	954

Item	Value
Total sources (slugs)	85
Phase B collection	cjp_columns_prod (new collection at 1536-dim — no in-place migration)

Per-chunk metadata schema

source_url	e.g. "https://opinion.inquirer.net/167388/..." or "source_materials/centenary_chapters/ch14-the-death-penalty.txt"
title	e.g. "Centenary, Ch.14: The Death Penalty"
publication_date	YYYY-MM-DD
bucket	"A" "B" "C" "D" "E"
bucket_name	human-readable, e.g. "Liberty and Rule of Law"
chunk_index	0-based within source
chunk_count	total chunks for this source
word_count	of this chunk
citation_safe	bool — Jacob's C5 review flag
ingested_at	ISO8601
embedder	"minilm" (will be "openai" post-cutover)

9. Retrieval layer

Item	Value
Module	backend/retrieval.py — retrieve(query, top_k, bucket, safe_only)
Top-k	5 (env: RETRIEVAL_TOP_K)
Distance threshold	0.55 (env: RETRIEVAL_DISTANCE_THRESHOLD; smoke-tested for MiniLM)
Diversity guardrail	max 2 chunks per source (env: RETRIEVAL_MAX_PER_SOURCE)
Bucket filter	optional, used for thematic biasing (not in default flow)
Citation-safe filter	via RETRIEVAL_SAFE_ONLY env (off in dev; will be true in prod)
Over-fetch	n_results = top_k * 3 to give the diversity filter room
Returns	List of RetrievedChunk dataclasses (text, title, date, url, bucket, distance, citation_safe)

10. Orchestrator (the per-turn flow)

answer_question(text, history) — backend/orchestrator.py

```

start_timer()

chunks = retrieve(text)

if not chunks:
    - pick a random fallback variant
    - return Response(fallback=True, ~30 ms latency, no LLM call)

build messages = [
    {role: system, content: persona_prompt},
    {role: system, content: "Provided sources: ..." + chunk texts},
    *history,
    {role: user, content: text},
]

answer = chat(messages) # -> Groq llama-3.3-70b-versatile
answer = trim_to_150_words(answer)

```

```

citations = [{title, date, url, bucket, distance, citation_safe} for c in chunks]

return Response(text=answer, citations=citations, latency_ms=..., fallback=False)

```

11. Robot adapter (future-proofing)

Item	Value
Module	backend/adapters.py
Protocol	RobotAdapter (speak / head_move / gaze_track / thinking_motion / handle_interrupt)
Live impl	WebAdapter — speak() does st.write + st.audio autoplay; others are no-ops
Future impl	ReachyAdapter — drops in late July / Aug, no other code changes needed

12. Knowledge base composition

Bucket	Theme	# Sources	# Chunks
A	Liberty and Rule of Law	44 (24 cols + 20 chapters)	~770
B	Prosperity and Economic Philosophy	7 columns	~25
C	Biographical and Personal	17 columns	~75
D	FLP Mission and Foundation	10 columns	~40
E	Signature Current Events Commentary	7 columns	~30
Total		85 sources	954 chunks

Citation-safe flag: 5 columns flagged citation_safe=False (Duterte ICC, Marcos ICC options, Trump defeat, ICC DOJ-OSG, ICC temporal-strategic) — pending Jacob's C5 review.

13. Ingestion pipeline

Run via python ingest_columns.py. 5 stages, all in ingest_columns.py:

```

Stage 1 - Load
+- if URL -> fetch HTML (browser-shaped headers + canonical URL guard)
|   -> trafilatura -> markdown
+- if file -> read .docx (Docx2txtLoader) or .txt/.md (raw read) -> markdown
    v
Stage 2 - Chunk
+- RecursiveCharacterTextSplitter, chunk_size=500 tokens, overlap=75 tokens
+- Tokenizer: cl100k_base (matches OpenAI for cutover compatibility)
+- Separators: ["\n\n", "\n", ". ", " ", ""]
    v
Stage 3 - Embed
+- all-MiniLM-L6-v2, batch_size=32, -> 384-dim vector per chunk
    v
Stage 4 - Upsert
+- ChromaDB cjp_columns_dev, deterministic chunk IDs ({slug}_c{NN})
    v
Stage 5 - Smoke test (optional, --smoke-test flag)
+- Sample query -> top-5 -> eyeball

```

Word-count guards: <400 words from any source raises (catches truncation / wrong-block extraction). The canonical-URL guard catches Inquirer's silent redirect to the section homepage when a column URL is dead.

Book chapter split: The 2001 book “A Centenary of Justice” was split into 20 chapter .txt files via scripts/extract_chapters_from_pdf.py (sequential title-search parser over the source PDF) before ingestion.

14. Configuration files

File	Purpose	Tracked?
.env	Real config — has Groq key	GITIGNORED
.env.example	Template, no real keys	tracked
.streamlit/config.toml	Disable file watcher, telemetry	tracked
.gitignore	excludes .venv/, chroma_store/, kb/, models/, source_materials/, *.log	tracked
requirements.txt	Pinned deps (~150 packages)	tracked
prompts/instructions.txt	Persona prompt	tracked
prompts/fallback.txt	Fallback variants (5)	tracked

15. Environment variables

Var	Current value	What it controls
GROQ_API_KEY	(your key)	LLM auth
GROQ_MODEL	llama-3.3-70b-versatile	LLM model
WHISPER_MODEL	small (just bumped)	STT model size
TTS_VOICE	en-US-AndrewNeural	TTS voice
RETRIEVAL_TOP_K	(default 5)	retrieval breadth
RETRIEVAL_DISTANCE_THRESHOLD	(default 0.55)	confidence cutoff
RETRIEVAL_MAX_PER_SOURCE	(default 2)	diversity guardrail
RETRIEVAL_SAFE_ONLY	(default false in dev)	citation-safe filter
CHROMA_COLLECTION	(default cjp_columns_dev)	ChromaDB collection
CONVERSATION_WINDOW_MINUTES	5 (not yet wired)	Step 1.9
TRIGGER_WORDS	that's enough,... (not yet wired)	Step 1.9
EMBEDDING_MODEL / EMBEDDING_DNEAD CODE		no longer read; needs cleanup

16. Phase A → Phase B (OpenAI cutover) checklist

Component	Phase A (now)	Phase B (after May 14 + audit)
LLM	Groq llama-3.3-70b-versatile	OpenAI gpt-4o

Component	Phase A (now)	Phase B (after May 14 + audit)
Embeddings	sentence-transformers all-MiniLM-L6-v2 (384-dim)	OpenAI text-embedding-3-small (1536-dim)
STT	faster-whisper small local	TBD — open: cloud-hosted faster-whisper, OpenAI Whisper API, or stay
TTS	edge-tts (free cloud)	TBD — possibly OpenAI TTS, possibly stay edge-tts
Vector store	ChromaDB cjp_columns_dev (laptop)	ChromaDB cjp_columns_prod on R-Pi (full re-embed required)
Backend host	Streamlit imports orchestrator	Possibly FastAPI behind HTTP, on R-Pi

17. What's NOT yet wired (Track 1 leftovers)

Step 1.9 — trigger-word interrupt (“thank you” / “next question” → reset state) and the 5-min conversation window auto-clear.

Step 1.10 — structured JSON logging to `logs/turns.jsonl` (per-turn input, citations, latency, fallback reason — required per HANDOVER §3 rule #13).

18. Known imperfections to revisit

Issue	Severity	Plan
EMBEDDING_MODEL / EMBEDDING_DIM envars are dead code	Critical	Remove next push
Ch 2 of book is 27,556 words (photo-plate captions absorbed)	Medium	Bring to next strategy session for chunking-of-oversized-chapters
Ch 18 / 19 boundary off by ~1k words	Low	Same
Ch 20 absorbs back matter (index, etc.)	Low-medium	Same
LLM occasionally ignores source-grounded rule on (RAG context) suppression of this tuning dry-runs	Low	Watch this tuning
Distance threshold 0.55 was a smoke-test guess for (working in practice)	Low	Re-tune at OpenAI cutover

End of pipeline review. For per-push history and recent decisions, see `PROGRESS.md` in the repo. For the strategic spec, see `CLAUDE.md` and `docs/handover_v2.md`.